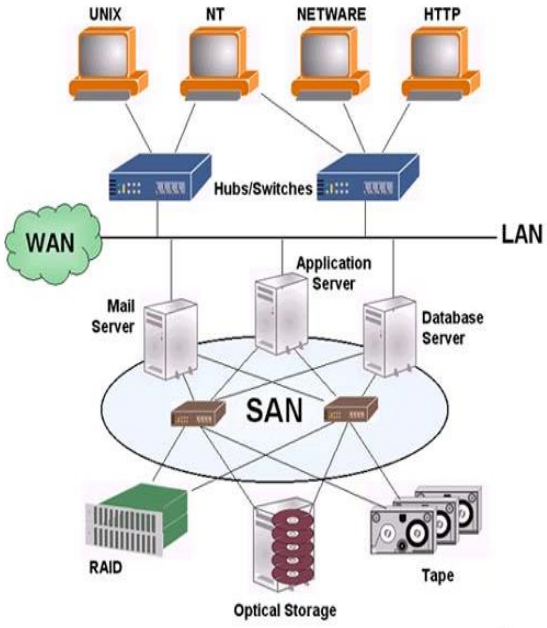


CLOUD COMPUTING COURSE

Lecture20- Cloud Storage (3)

ABSTRACT
Lecture20 Introduces SAN storage.

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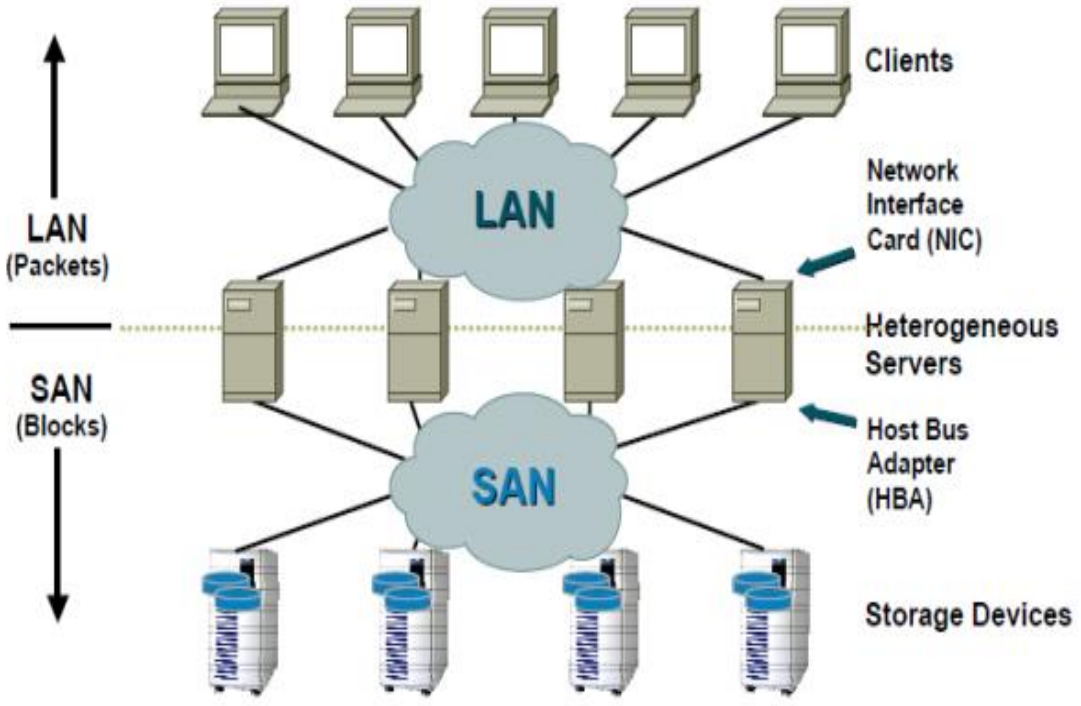
Centralized Storage

3. SAN (Storage Area Network):

Now, data has become the most valuable asset since the Internet and e-commerce are growing at an explosive rate. How to store, protect and manage data in an effective way is a big challenge for IT technicians. Here, SAN, storage area network, has been developed to provide a creative model for data storing in the data center.

SAN Definition: is a specialized, high-speed network that provides block-level network access to storage.

SAN Component: SANs are typically composed of **hosts, switches** and **storage devices** that are interconnected using a variety of technologies, topologies, and protocols.



SAN Is a Dedicated Network for Attaching Servers to Storage Devices

SAN Access Mechanism

SAN makes a network of storage **devices accessible to multiple servers.**

SAN devices **appear to servers as locally attached drives,** eliminating traditional network bottlenecks.

A SAN is block-based storage, leveraging a high-speed architecture that connects servers to their **logical disk units (LUNs**).

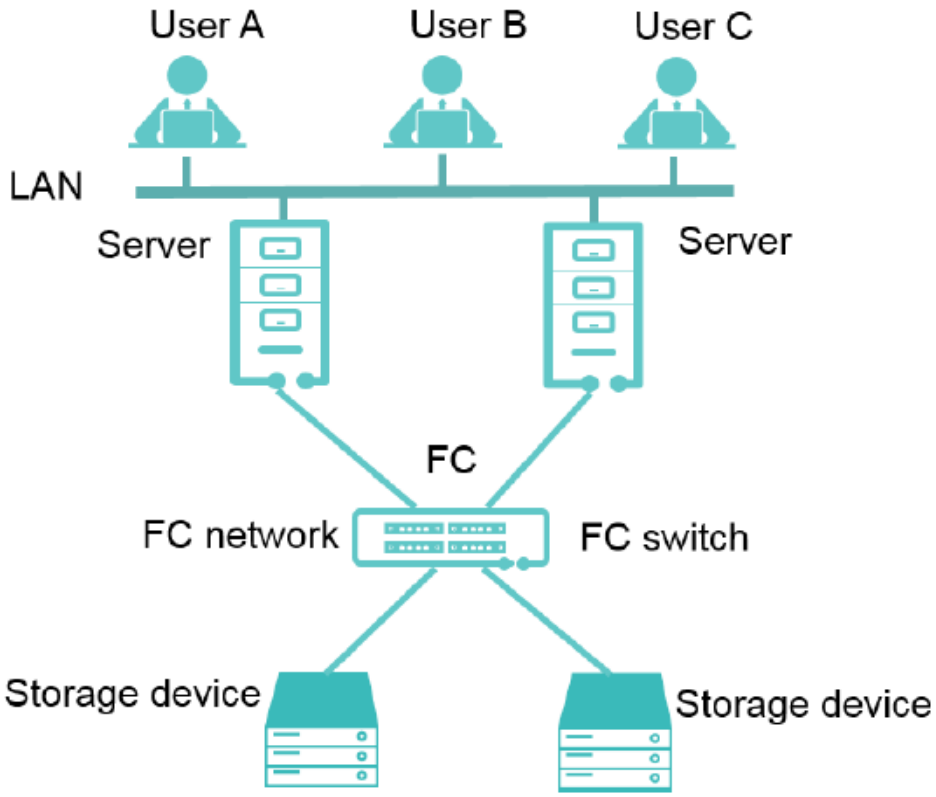
A LUN is a range of blocks provisioned from a pool of shared storage and presented to the server **as a logical disk.**

The **server partitions and formats those blocks,** typically with a file system, so that it can store data on the LUN just as it would on local disk storage.

SAN Types

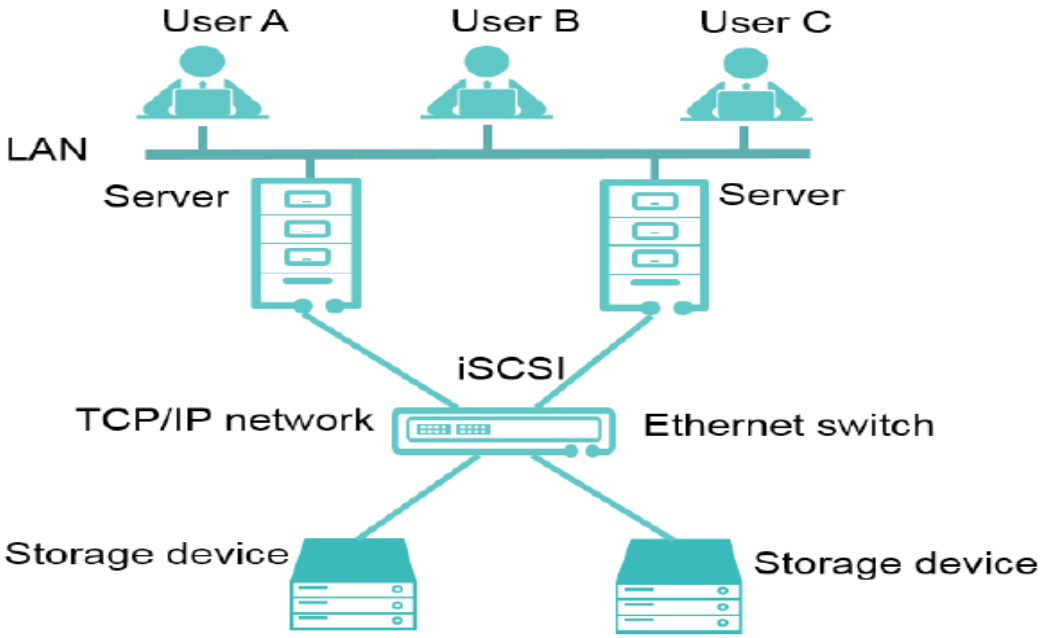
SAN solutions are available as many types, the most common two types are:

- **FC-SAN (Fiber Channel SAN):** Storage and servers are connected via a high-speed network of interconnected fiber channel switches. This is used for mission-critical applications where uninterrupted data access is required.



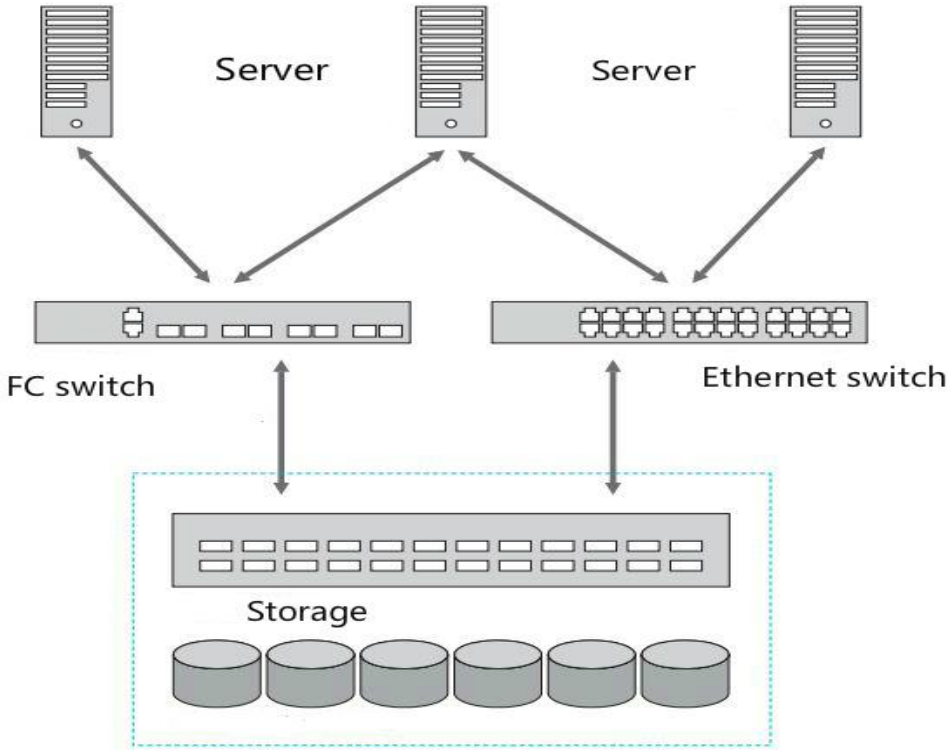
FC SAN Network

- **Internet Small Computer System Interface (iSCSI) Protocol/IP SAN:** This infrastructure gives the flexibility of a low-cost IP network. It uses iSCSI protocol, a transmission standard over TCP/IP, to transfer block data over an Ethernet network. IP SAN allows different servers to access pools of the shared block storage devices by storage protocols.



IP SAN network

Composite Solution (both FC SAN and IP SAN)



IP SAN vs FC SAN: What Is the Difference?

Here is a table shows the differences of IP SAN vs FC SAN.

	IP SAN	FC SAN
Network	Gigabit Ethernet network	fiber channel network
Cost	Low	High
Transmission speed	Fast	Extremely fast
Transmission protocol	TCP/IP	fiber channel
Extensibility	More easier to extend	Easy to extend

SAN Use Cases

Storage area networks are frequently deployed in support of business-critical, performance-sensitive applications such as:

- **Oracle databases.** These are frequently business-critical and require the highest performance and availability.
- **Microsoft SQL Server databases.** Like Oracle databases, MS SQL Server databases commonly store an enterprise's most valuable data, so they require the highest performance and availability.
- **Large virtualization deployments using VMware, KVM, or Microsoft Hyper-V.** These environments often extend to thousands of virtual machines running a broad range of operating systems and applications, with different performance requirements. Virtualized environments concentrate many applications, so infrastructure reliability becomes even more important because a failure can cause multiple application outages.
- **Large virtual desktop infrastructures (VDIs).** These environments serve virtual desktops to large numbers of an organization's users. Some VDI environments can easily number in the tens of thousands of virtual desktops. By centralizing the virtual desktops, organizations can more easily manage data protection and data security.
- **SAP or other large ERP or CRM environments.** SAN architectures are ideal for enterprise resource planning and customer relationship management workloads.

SAN vs. NAS

	NAS	SAN
Main Target	Are methods of managing storage centrally and sharing that storage with multiple hosts (servers) .	
Network	Ethernet-based	Ethernet and Fibre Channel
Focus on	Ease of use, manageability, scalability, and lower total cost of ownership (TCO).	high performance and low latency
Partitioning	NAS **storage controllers** partition the storage and then own the file system	The **server** partitions and formats SAN storage blocks with its file system
Protocols	CIFS, NFS	FCP, iSCSI Protocol